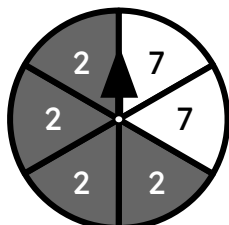




Comparing multiple representations to calculate theoretical probabilities for combined events

Task A: Showing two-stage trials across different diagrams



spinner

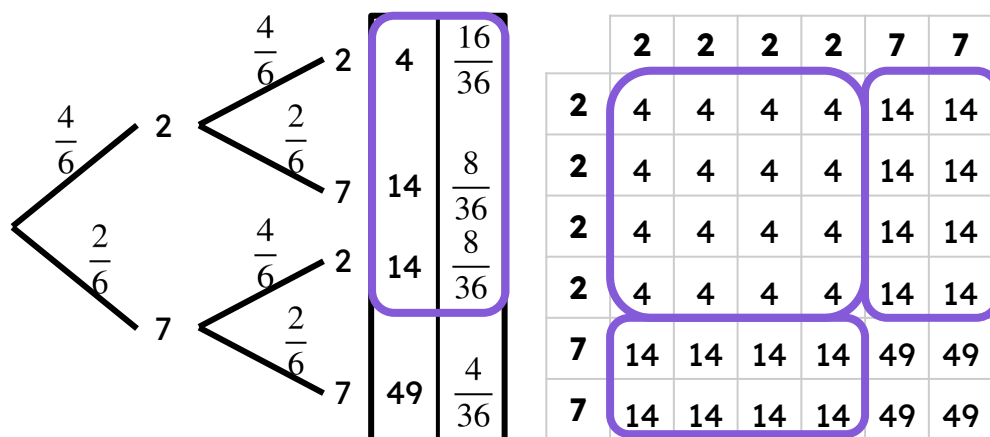
1) Trial: this spinner is spun twice, with the outcomes of each spin multiplied together.

a) Complete the outcome table and probability tree for this trial.

b) Circle the probabilities and outcomes for the event "even number".

c) Find $P(\text{multiple of } 7)$

$$\frac{20}{36}$$



2) Trial: spinner A and B will be spun once each. The outcomes are added together.

		spinner A							
		1	1	1	2	2	3	3	3
spinner B	4	5	5	5	6	6	7	7	7
	5	6	6	6	7	7	8	8	8
	5	6	6	6	7	7	8	8	8

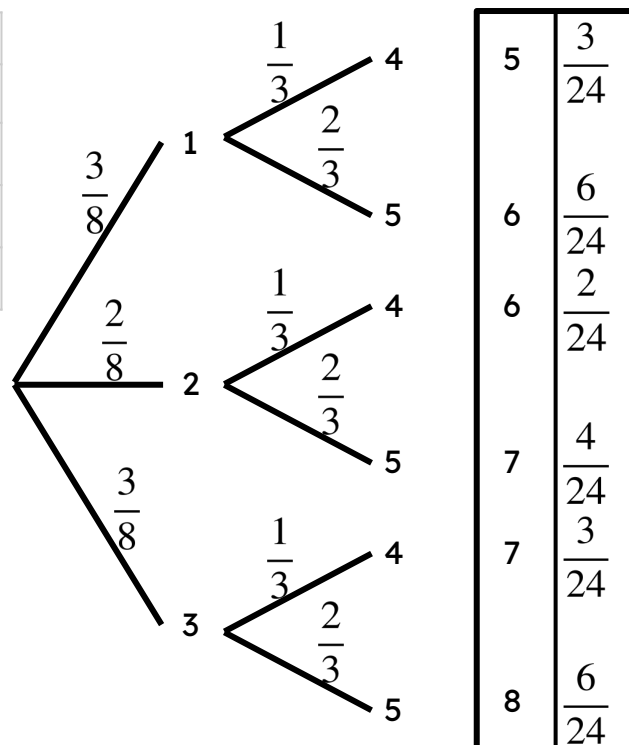
a) Use the outcome table to complete the probability tree.

b) How many prime outcomes are there?

10 prime outcomes

c) Find $P(6 \text{ or } 8)$ using the probability tree.

$$\frac{6}{24} + \frac{2}{24} + \frac{6}{24} = \frac{14}{24}$$



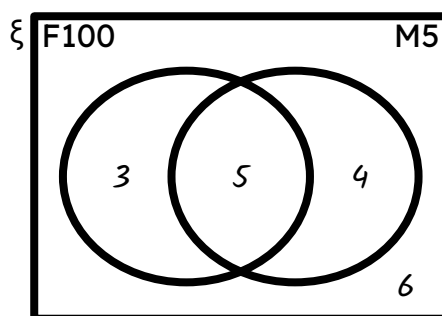
Task B: Statements for events across different diagrams

1) Trial: a spinner is spun once.

Events: the spinner lands on a factor of 100 (F100) or a multiple of 5 (M5).

a) Use the outcome table to complete the Venn diagram showing the number of outcomes for each event.

	(M5)	not M5
F100	5 20 25 10 50	1 2 4
not F100	15 35 75 1005	3 7 11 16 21 33



Find these probabilities:

b) $P(\text{M5 and F100}) = \frac{5}{18}$

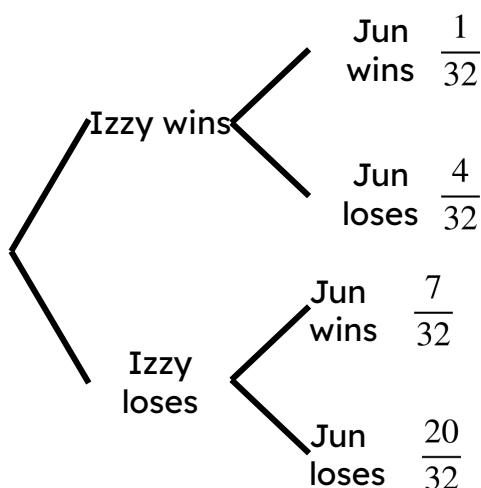
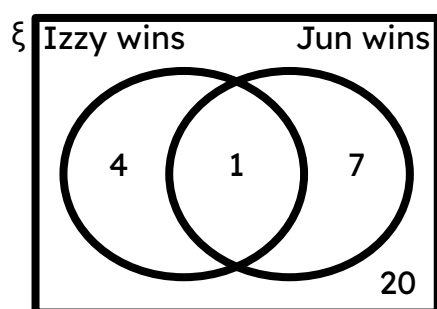
c) $P(\text{M5}) = \frac{9}{18}$

d) $P(\text{M5 or F100}) = \frac{12}{18}$

2) Trial: selecting a random number from a game at a fair.

Events: Izzy wins, Jun wins

a) Use the Venn diagram showing the number of outcomes that result in Izzy or Jun winning to complete the probability tree.



Find these probabilities:

b) $P(\text{Izzy wins}) = \frac{5}{32}$

c) $P(\text{only Jun wins}) = \frac{7}{32}$

d) $P(\text{either win}) = \frac{12}{32}$

3) Trial: a spinner is spun once.

Events: the spinner lands on a square number or an even number.

a) Complete both the probability tree with probabilities and events, and complete the outcome table with one possible missing outcome.

	square	not square	
even	0 4 16	2 6 10	even
	36 84 100	22 38 60	
	144 196		
odd	1 9	3 5 13	odd
	25 49	21 35 45	
		57 69 80	

	square	$\frac{4}{14}$
	not square	$\frac{3}{14}$
	square	$\frac{2}{14}$
	not square	$\frac{5}{14}$

The missing outcome was an odd number that wasn't square.

For example:

7, 11, 15, 17, 19, 23, ...

Find:

b) $P(\text{square}) = \frac{12}{28} = \frac{6}{14}$

c) $P(\text{either square or even}) = \frac{18}{28} = \frac{9}{14}$

d) $P(\text{either square or even, but not both}) = \frac{10}{28} = \frac{5}{14}$

